

2.2.2 Block Diagram Representation of Discrete-Time Systems

An adder.

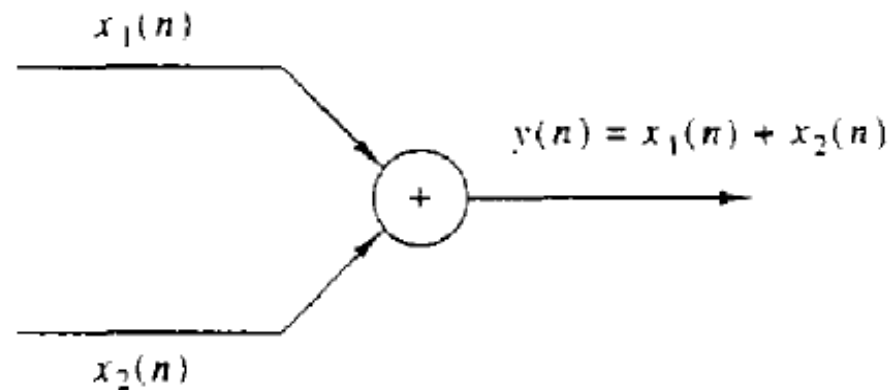


Figure 2.13 Graphical representation of an adder.

A constant multiplier.



Figure 2.14 Graphical representation of a constant multiplier.

A signal multiplier.

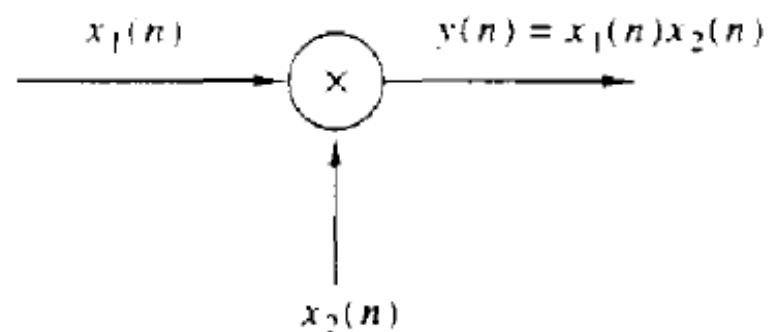


Figure 2.15 Graphical representation of a signal multiplier.

A unit delay element.

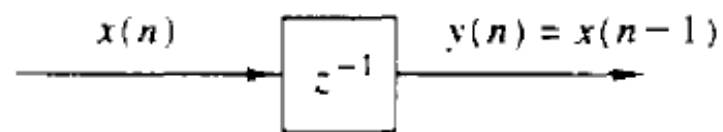


Figure 2.16 Graphical representation of the unit delay element.

Thus this basic building block requires memory. The use of the symbol z^{-1} to denote the unit of delay will become apparent when we discuss the z -transform in Chapter 3.

A unit advance element.



Figure 2.17 Graphical representation of the unit advance element.

We observe that any such advance is physically impossible in real time, since, in fact, it involves looking into the future of the signal. On the other hand, if we store the signal in the memory of the computer, we can recall any sample at any time. In such a nonreal-time application, it is possible to advance the signal $x(n)$ in time.

Example 2.2.3

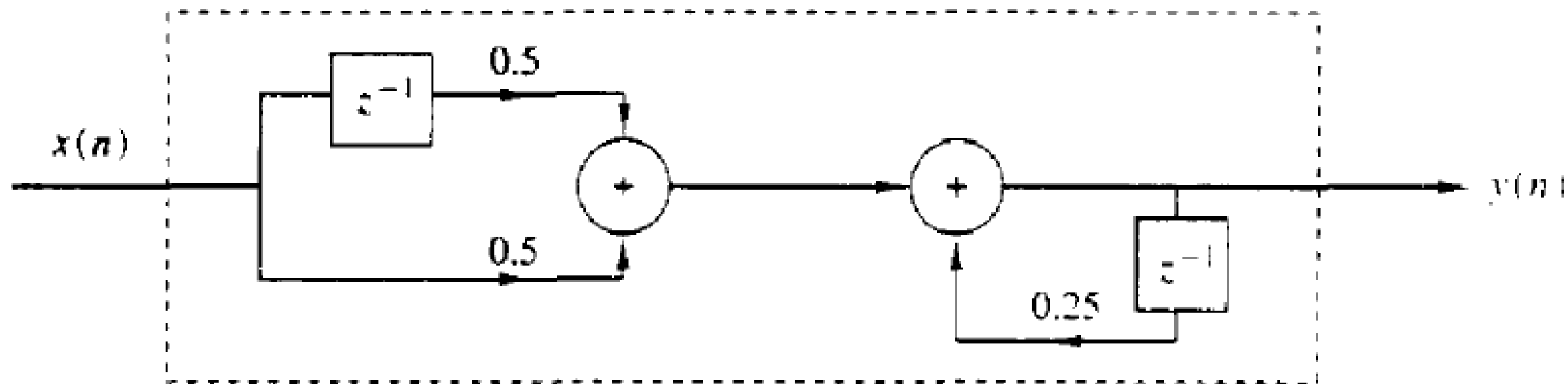
Using basic building blocks introduced above, sketch the block diagram representation of the discrete-time system described by the input-output relation.

$$y(n) = \frac{1}{4}y(n-1) + \frac{1}{2}x(n) + \frac{1}{2}x(n-1) \quad (2.2.5)$$

where $x(n)$ is the input and $y(n)$ is the output of the system.

$$y(n] = \frac{1}{4}y[n - 1] + \frac{1}{2}x[n] + \frac{1}{2}x[n - 1]$$

Black box



(a)

2.2.3 Classification of Discrete-Time Systems

Static versus dynamic systems. A discrete-time system is called *static* or *memoryless* if its output at any instant n depends at most on the input sample at the same time, but not on past or future samples of the input. In any other case, the system is said to be *dynamic* or to have *memory*. If the output of a system at time n is completely determined by the input samples in the interval from $n - N$ to n ($N \geq 0$), the system is said to have *memory* of duration N . If $N = 0$, the system is *static*. If $0 < N < \infty$, the system is said to have *finite memory*, whereas if $N = \infty$, the system is said to have *infinite memory*.

The systems described by the following input–output equations

$$y(n) = ax(n) \quad (2.2.7)$$

$$y(n) = nx(n) + bx^3(n) \quad (2.2.8)$$

are both static or memoryless. Note that there is no need to store any of the past inputs or outputs in order to compute the present output. On the other hand, the systems described by the following input–output relations

$$y(n) = x(n) + 3x(n - 1) \quad (2.2.9)$$

$$y(n) = \sum_{k=0}^n x(n - k) \quad (2.2.10)$$

$$y(n) = \sum_{k=0}^{\infty} x(n - k) \quad (2.2.11)$$

are dynamic systems or systems with memory.

Time-invariant versus time-variant systems. We can subdivide the general class of systems into the two broad categories, time-invariant systems and time-variant systems. A system is called time-invariant if its input-output characteristics do not change with time. To elaborate, suppose that we have a system $\mathcal{T}$ in a relaxed state which, when excited by an input signal $x(n)$, produces an output signal $y(n)$. Thus we write

$$y(n) = \mathcal{T}[x(n)] \quad (2.2.13)$$

Now suppose that the same input signal is delayed by k units of time to yield $x(n - k)$, and again applied to the same system. If the characteristics of the system do not change with time, the output of the relaxed system will be $y(n - k)$. That is, the output will be the same as the response to $x(n)$, except that it will be delayed by the same k units in time that the input was delayed. This leads us to define a time-invariant or shift-invariant system as follows.

Definition. A relaxed system $\mathcal{T}$ is *time invariant* or *shift invariant* if and only if

$$x(n) \xrightarrow{\mathcal{T}} y(n)$$

implies that

$$x(n - k) \xrightarrow{\mathcal{T}} y(n - k) \tag{2.2.14}$$

for every input signal $x(n)$ and every time shift k .

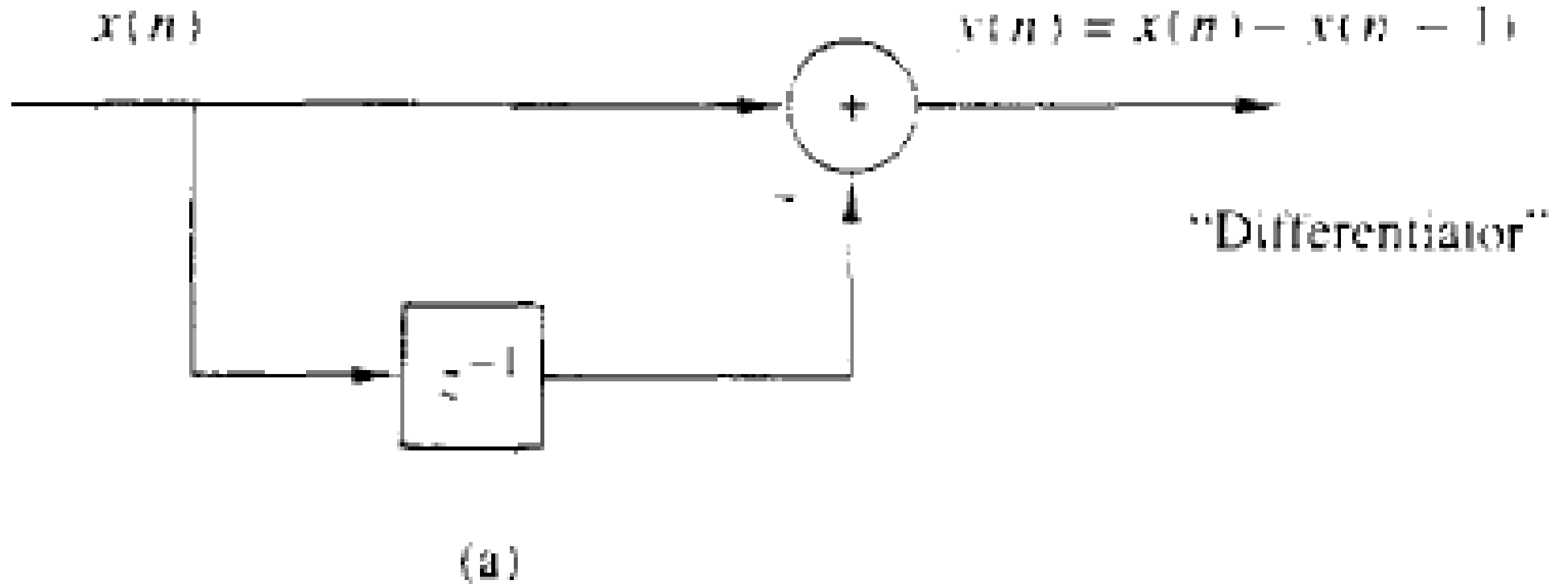
To determine if any given system is time invariant, we need to perform the test specified by the preceding definition. Basically, we excite the system with an arbitrary input sequence $x(n)$, which produces an output denoted as $y(n)$. Next we delay the input sequence by same amount k and recompute the output. In general, we can write the output as

$$y(n, k) = T[x(n - k)]$$

Now if this output $y(n, k) = y(n - k)$, for all possible values of k , the system is time invariant. On the other hand, if the output $y(n, k) \neq y(n - k)$, even for one value of k , the system is time variant.

Example 2.2.4

Determine if the systems shown in Fig. 2.19 are time invariant or time variant.



Example 2.2.4

$$y(n) = x(n) - x(n-1]$$

Let us assume the signal of the form

$$y_1(n) = x_1(n) - x_1(n-1] \quad \text{--- (i)}$$

Let us introduce time delay 'k' in the input signal in eq (i), then

$$x_2(n) = x_1(n-k)$$

The delay input therefore results in the output:

$$y_2(n) = x_2(n) - x_2(n-1]$$

$$y_2(n) = x_1(n-k) - x_1(n-k-1] \quad \text{--- (ii)}$$

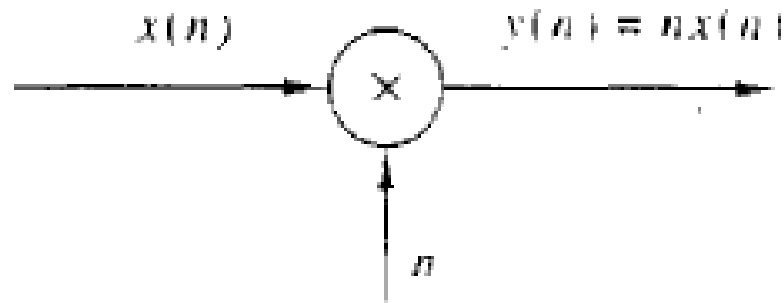
Let us introduce the same time delay 'k' in the output of the eq, i.e.

$$y_1(n-k) = x_1(n-k) - x_1(n-k-1] \quad \text{--- (iii)}$$

As (ii) & (iii) are same:

$$y_2(n) = y_1(n-k)$$

Hence, system is time-invariant.



(b)

Example 2.2.4 (b)

$$y(n) = nx(n)$$

Let

$$y_1(n) = nx_1(n)$$

Introduce time delay 'k' in input,

$$x_2(n) = x_1(n-k)$$

Hence,

$$y_2(n) = nx_2(n) = nx_1(n-k) \quad \text{--- (i)}$$

Introduce time delay 'k' in the output,

$$y_1(n-k) = (n-k)x_1(n-k)$$

$$y_1(n-k) = nx_1(n-k) - kx_1(n-k) \quad \text{--- (ii)}$$

As

$$y_2(n) \neq y_1(n-k)$$

Therefore, this system is time variant.



"Folder"

(c)

Example 2.2.4(c)

$$y(n) = x(-n)$$

Let $y_1(n) = x_1(-n)$

Introduce time delay 'k' in the input:

$$x_2(n) = x_1(n-k)$$

Hence, $y_2(n) = x_2(-n) = x_1(-n-k)$ — (i)

Introduce time delay 'k' in the output:

$$y_1(n-k) = x_1(-(n-k)) = x_1(-n+k)$$
 — (ii)

$$\text{As } y_2(n) \neq y_1(n-k)$$

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 Hence, this system is time-variant.

Linear versus nonlinear systems. The general class of systems can also be subdivided into linear systems and nonlinear systems. A linear system is one that satisfies the *superposition principle*. Simply stated, the principle of superposition requires that the response of the system to a weighted sum of signals be equal to the corresponding weighted sum of the responses (outputs) of the system to each of the individual input signals. Hence we have the following definition of linearity.

Definition. A relaxed $\mathcal{T}$ system is linear if and only if

$$\mathcal{T}[a_1x_1(n) + a_2x_2(n)] = a_1\mathcal{T}[x_1(n)] + a_2\mathcal{T}[x_2(n)] \quad (2.2.26)$$

for any arbitrary input sequences $x_1(n)$ and $x_2(n)$, and any arbitrary constants a_1 and a_2 .

Figure 2.20 gives a pictorial illustration of the superposition principle.

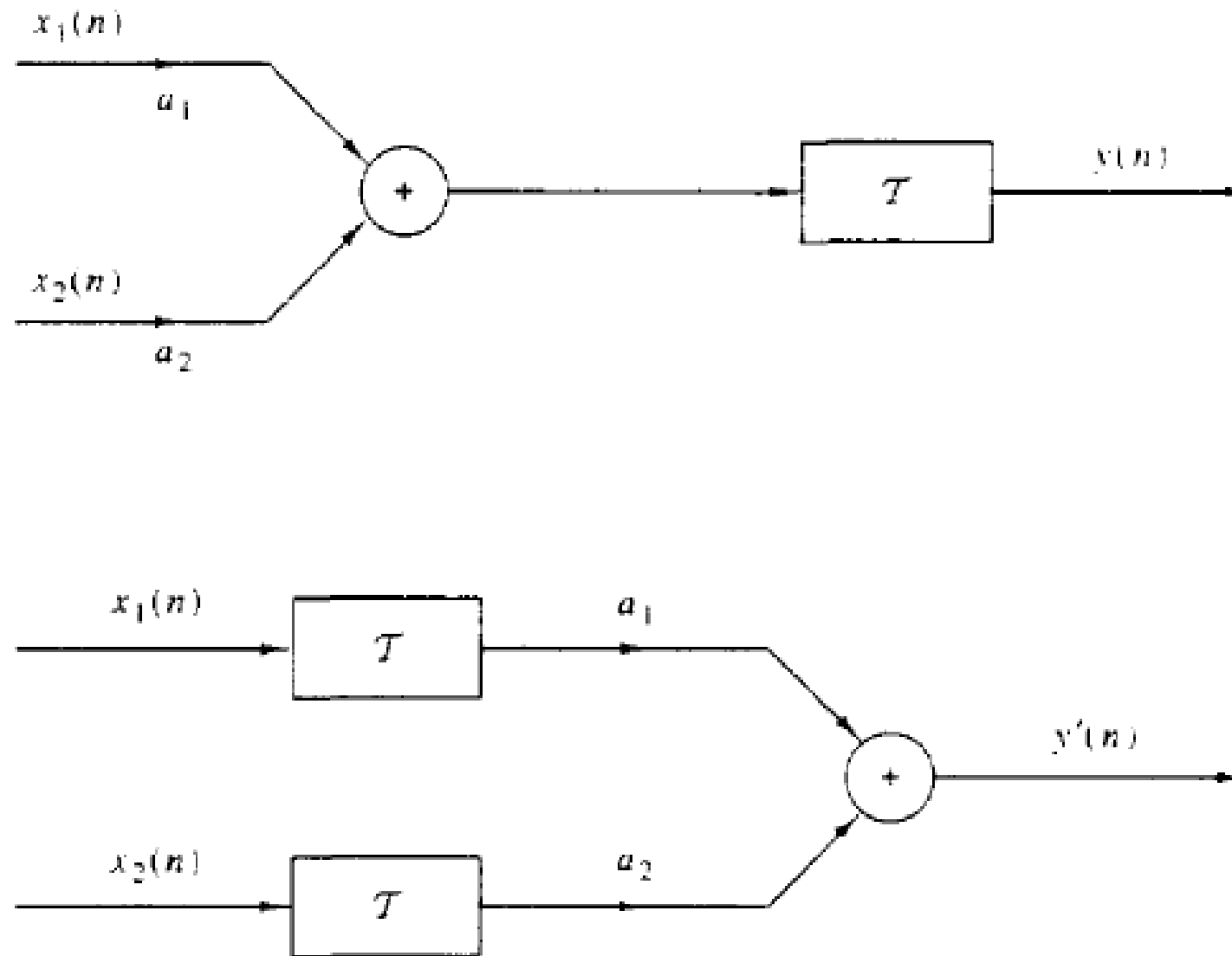


Figure 2.20 Graphical representation of the superposition principle. $\mathcal{T}$ is linear if and only if $y(n) = y'(n)$.

The superposition principle embodied in the relation (2.2.26) can be separated into two parts. First, suppose that $a_2 = 0$. Then (2.2.26) reduces to

$$\mathcal{T}[a_1 x_1(n)] = a_1 \mathcal{T}[x_1(n)] = a_1 y_1(n) \quad (2.2.27)$$

where

$$y_1(n) = \mathcal{T}[x_1(n)]$$

The relation (2.2.27) demonstrates the *multiplicative* or *scaling property* of a linear system. That is, if the response of the system to the input $x_1(n)$ is $y_1(n)$, the response to $a_1 x_1(n)$ is simply $a_1 y_1(n)$. Thus any scaling of the input results in an identical scaling of the corresponding output.

Second, suppose that $a_1 = a_2 = 1$ in (2.2.26). Then

$$\begin{aligned} \mathcal{T}[x_1(n) + x_2(n)] &= \mathcal{T}[x_1(n)] + \mathcal{T}[x_2(n)] \\ &= y_1(n) + y_2(n) \end{aligned} \quad (2.2.28)$$

This relation demonstrates the *additivity property* of a linear system. The additivity and multiplicative properties constitute the superposition principle as it applies to linear systems.

Example 2.2.5

Determine if the systems described by the following input–output equations are linear or nonlinear.

$$\text{(a)} \quad y(n) = nx(n) \qquad \text{(b)} \quad y(n) = x(n^2) \qquad \text{(c)} \quad y(n) = x^2(n)$$

$$\text{(d)} \quad y(n) = Ax(n) + B \qquad \text{(e)} \quad y(n) = e^{x(n)}$$

Solution

(a) For two input sequences $x_1(n)$ and $x_2(n)$, the corresponding outputs are

$$\begin{aligned} y_1(n) &= nx_1(n) \\ y_2(n) &= nx_2(n) \end{aligned} \tag{2.2.31}$$

A linear combination of the two input sequences results in the output

$$\begin{aligned} y_3(n) &= \mathcal{T}\{a_1x_1(n) + a_2x_2(n)\} = n[a_1x_1(n) + a_2x_2(n)] \\ &= a_1nx_1(n) + a_2nx_2(n) \end{aligned} \tag{2.2.32}$$

On the other hand, a linear combination of the two outputs in (2.2.31) results in the output

$$a_1y_1(n) + a_2y_2(n) = a_1nx_1(n) + a_2nx_2(n) \tag{2.2.33}$$

Since the right-hand sides of (2.2.32) and (2.2.33) are identical, the system is linear.

(b) As in part (a), we find the response of the system to two separate input signals $x_1(n)$ and $x_2(n)$. The result is

$$\begin{aligned}y_1(n) &= x_1(n^2) \\y_2(n) &= x_2(n^2)\end{aligned}\tag{2.2.34}$$

The output of the system to a linear combination of $x_1(n)$ and $x_2(n)$ is

$$y_3(n) = \mathcal{T}[a_1x_1(n) + a_2x_2(n)] = a_1x_1(n^2) + a_2x_2(n^2)\tag{2.2.35}$$

Finally, a linear combination of the two outputs in (2.2.34) yields

$$a_1y_1(n) + a_2y_2(n) = a_1x_1(n^2) + a_2x_2(n^2)\tag{2.2.36}$$

By comparing (2.2.35) with (2.2.36), we conclude that the system is linear.

(c) The output of the system is the square of the input. (Electronic devices that have such an input–output characteristic and are called square-law devices.) From our previous discussion it is clear that such a system is memoryless. We now illustrate that this system is nonlinear.

The responses of the system to two separate input signals are

$$\begin{aligned}y_1(n) &= x_1^2(n) \\ y_2(n) &= x_2^2(n)\end{aligned}\tag{2.2.37}$$

The response of the system to a linear combination of these two input signals is

$$\begin{aligned}y_3(n) &= \mathcal{T}[a_1x_1(n) + a_2x_2(n)] \\ &= [a_1x_1(n) + a_2x_2(n)]^2 \\ &= a_1^2x_1^2(n) + 2a_1a_2x_1(n)x_2(n) + a_2^2x_2^2(n)\end{aligned}\tag{2.2.38}$$

On the other hand, if the system is linear, it would produce a linear combination of the two outputs in (2.2.37), namely,

$$a_1y_1(n) + a_2y_2(n) = a_1x_1^2(n) + a_2x_2^2(n)\tag{2.2.39}$$

Since the actual output of the system, as given by (2.2.38), is not equal to (2.2.39), the system is nonlinear.

(d) Assuming that the system is excited by $x_1(n)$ and $x_2(n)$ separately, we obtain the corresponding outputs

$$\begin{aligned}y_1(n) &= Ax_1(n) + B \\y_2(n) &= Ax_2(n) + B\end{aligned}\tag{2.2.40}$$

A linear combination of $x_1(n)$ and $x_2(n)$ produces the output

$$\begin{aligned}y_3(n) &= \mathcal{T}[a_1x_1(n) + a_2x_2(n)] \\&= A[a_1x_1(n) + a_2x_2(n)] + B \\&= Aa_1x_1(n) + a_2Ax_2(n) + B\end{aligned}\tag{2.2.41}$$

On the other hand, if the system were linear, its output to the linear combination of $x_1(n)$ and $x_2(n)$ would be a linear combination of $y_1(n)$ and $y_2(n)$, that is,

$$a_1y_1(n) + a_2y_2(n) = a_1Ax_1(n) + a_1B + a_2Ax_2(n) + a_2B\tag{2.2.42}$$

Clearly, (2.2.41) and (2.2.42) are different and hence the system fails to satisfy the linearity test.

(e) Note that the system described by the input–output equation

$$y(n) = e^{x(n)} \quad (2.2.43)$$

is relaxed. If $x(n) = 0$, we find that $y(n) = 1$. This is an indication that the system is nonlinear. This, in fact, is the conclusion reached when the linearity test, is applied.